1 informal proofs

proofs are where we can use what we have learned so far with propositional equivalence and rules of inference. in this chapter we cover basic fundamental proof techniques and strategies to prove mathematical theorems, in particular to those stated as implications.

1.1 direct proof

for a direct proof, we assume that the premise p is true and then show that the conclusion q must follow.

1.1.1 example

Theorem: if n is an odd integer, then n^3 is an odd integer.
Proof:

1. **assume p is true**: assume n is an odd integer. this means n can be written as $2k + 1$ for some integer k.
2. **show q must be true**: substitute $(2k + 1)$ for n in n^3:
   - $n^3 = (2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1$
3. **conclude**: after factoring the expression on the right, we get $n^3 = 2(4k^3 + 6k^2 + 3k) + 1$, and since k is an integer, $4k^3 + 6k^2 + 3k$ is just another big integer, and we can call it m. we can replace that in the factored equation, which gives us $n^3 = 2m + 1$ – the definition of an odd integer. ■

1.2 proof by contraposition

proof by contraposition is a type of indirect proof, where we prove the *contrapositive* of an implication, which is $\neg q \to \neg p$ – logically equivalent to $p \to q$.

1.3 proof by contradiction

1.4 which to choose